

Handover Note - Technical Documentation & Status Report

🌐 Deployment Infrastructure (Temporary Test Environment)

Status: Deployed on free/dev tiers. **Expiration Risk: High (30 days).**

Component	Service Provider	URL / Connection
Frontend	Vercel	https://loaninneed.vercel.app
Backend	Railway	https://loaninneed-backend-production-7028.up.railway.app
Database	Render (PostgreSQL)	Connection string via Env Vars
Storage	Supabase	S3 Compatible API

🔗 Technical Implementation Details

1. Referral System & Security (HMAC)

The referral tracking system uses a cryptographic signature to prevent tampering.

- **Algorithm:** HMAC-SHA256
- **Key Management:**
 - Each partner is assigned a unique 32-byte `secretKey` during registration.
 - This key is encrypted (AES-256) before storage in the PostgreSQL `Partner` table.
- **Link Generation:**
 - **Payload:** `${partnerId}|${timestamp}`
 - **Signature:** `HMAC_SHA256(payload, decryptedSecretKey)`
 - **URL Structure:** `/signup?pid=[ID]&ts=[TIMESTAMP]&sig=[HEX_SIGNATURE]`
- **Validation:**
 - When a user clicks the link, the backend re-computes the signature using the claimed `pid` and `ts`.
 - If the computed signature matches the `sig` param, the link is valid.

2. Attribution Flow (Frontend ↔ Backend)

- **Capture (Frontend):**
 - Component: `ReferralTracker.tsx`.
 - Logic: Detects `pid`, `ts`, `sig` in URL. Stores them in `localStorage` key `lin_attribution`.
 - Persistence: Data persists even if the user navigates away and comes back to register later.
- **Submission:**
 - During registration (`lib/api.ts`), the `lin_attribution` object is retrieved and injected into the user registration payload.
- **Processing (Backend):**
 - `registerUser` controller validates the signature again.

- Creates **User** record with **attributedPartnerId**.
- Logs the event in **AttributionLog** table for auditing.

3. API Endpoints Status

Partner Management

- **POST /api/partners/register** - **Functional** (Supports Individuals & Firms).
- **POST /api/partners/login** - **Functional** (Email/Password).
- **GET /api/partners/profile** - **Functional**.
- **GET /api/partners/link** - **Functional** (Generates signed production links).
- **GET /api/partners/earnings** - **Functional** (Returns detailed user/loan stats).

User Operations

- **POST /api/users/register** - **Functional** (Handles Attribution).
- **POST /api/auth/verify-otp** - **Mocked** (See Bypasses).

Core Services

- **POST /api/documents/upload** - **tuned** (Transaction timeout increased to 20s).

Critical Bypasses & Test Data (Must Read)

1. SMTP / Email System

- **Issue:** No SMTP server provided.
- **Bypass:** Registration emails are **NOT SENT**.
- **Recovery:** Partner credentials are returned in the API response (downloadable file on frontend) immediately after creation.

2. Verification APIs (OTP/KYC)

- **Issue:** No credits/integrations for SMS or Pan Verification APIs.
- **Bypass:**
 - **Mobile OTP:** Use Master OTP **261102** for ALL phone verifications.
 - **PAN/Aadhaar:** Accepts any valid format string; no external validation is performed.

Test Report Summary

Verified by: Automated Test Suite (**Backend/tester**) + Manual E2E Walkthrough.

- **Scenario 1: Firm Registration**
 - Input: **firmName="TechCorp"**, **gstNumber="29ABC..."**, **panNumber=null**.
 - Result: **PASS** (Firm Name saved & displayed).
- **Scenario 2: Referral Attribution**
 - Action: User clicks link **?pid=123...**. Registers.
 - Result: **PASS** (User linked to Partner 123).
- **Scenario 3: Earnings Calculation**

- Action: Attributed user applies for loan of ₹50,000.
 - Result: **PASS** (Partner dashboard shows ₹50,000 loan & pending commission).
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Generated by Antigravity - Jan 28, 2026